

One Law for the Radial Acceleration Relation and the External Field Effect: the de Sitter–Unruh Interpolation and $a_0 = c^2 \sqrt{\Lambda/32\pi}$

C. Zimmerman
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The *external field effect* (EFE) is the sharpest qualitative test separating modified gravity from particle dark matter: a uniform external field cannot touch a system’s internal dynamics under Newton+dark matter, but *must* in any non-linear modified-gravity law. This paper is the internally-consistent companion to an earlier treatment: there the isolated radial acceleration relation (RAR) used the framework’s de Sitter–Unruh interpolation $g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}} a_0}$ while the EFE was written with the unrelated “simple” MOND interpolation $\nu = \frac{1}{2} + \sqrt{\frac{1}{4} + 1/y}$. A modified-gravity theory has *one* interpolation function; here we use the framework’s own de Sitter–Unruh form $\nu(y) = \sqrt{1 + 1/y}$ for *both* the RAR and the EFE, at the framework’s own acceleration scale $a_0 = c^2 \sqrt{\Lambda/32\pi} = 9.36 \times 10^{-11} \text{ m s}^{-2}$. The qualitative physics is unchanged; the quantitative predictions sharpen and, honestly, *shrink*: because a_0 is lower than the canonical MOND value, the Milky Way’s external field sits at $g_{\text{ext}} = 2.30 a_0$ (not $1.8 a_0$), driving the system deeper into the external-field-suppressed regime. The wide-binary boost is $G_{\text{eff}}/G \simeq 1.14$ (orbit-averaged; +6.6% in velocity), *not* the 1.3 of the simple interpolation. We give the corrected figures, state the current data placement honestly (consistent with, but not confirmed against, present wide-binary measurements), and keep the acceleration scale’s coefficient an explicit, underived posit.

I. TWO EXPLANATIONS, ONE DECISIVE TEST

Spiral galaxies rotate too fast for their visible mass: below a characteristic acceleration the observed acceleration g_{obs} exceeds the Newtonian value g_{bar} from baryons alone, following the tight *radial acceleration relation* (RAR), with $g_{\text{obs}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$ at low accelerations [1]. Two pictures explain this. *Dark matter*: each galaxy sits in a halo of unseen particles. *Modified gravity* (MOND [2]): gravity itself is enhanced below a_0 , with no new matter. On rotation curves the two are nearly degenerate. The *external field effect* breaks the degeneracy cleanly, because it tests a *principle* rather than a fit: does a system’s internal gravity care about the gravitational field of its surroundings?

II. WHY DARK MATTER FORBIDS IT, AND MODIFIED INERTIA REQUIRES IT

Newton + dark matter: no effect. Newtonian gravity obeys the *strong equivalence principle*: a spatially uniform external field $\mathbf{g}_{\text{ext}}$ accelerates every part of a system equally and cancels from the internal equations of motion. A dwarf galaxy falling through a cluster has the *same* internal dynamics as one in deep space. Internal physics is blind to the external field.

The framework: an unavoidable effect. In this framework the modification is to *inertia*, arising from the de Sitter–Unruh temperature $T_{\text{eff}} = (\hbar/2\pi c k_B) \sqrt{a^2 + (cH)^2}$ a body reads off the accelerated vacuum. The response to a force depends on the *total* acceleration the body actually carries—internal + external—so a uniform $\mathbf{g}_{\text{ext}}$ shifts the operating point and changes the *internal* gravity. The strong equivalence

principle is violated; this is a *prediction*, not a defect. Quantitatively, the framework’s isolated law is the de Sitter–Unruh interpolation

$$g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}} a_0} \equiv \nu(g_{\text{bar}}) g_{\text{bar}}, \quad \nu(y) = \sqrt{1 + \frac{a_0}{y}}, \quad (1)$$

with the deep-MOND limit $g_{\text{obs}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$ and the Newtonian limit $g_{\text{obs}} \rightarrow g_{\text{bar}}$. The EFE is obtained from the *same* ν by the one-dimensional aligned-field construction

$$g_{\text{obs}} = \nu(g_N + g_{\text{ext}}) (g_N + g_{\text{ext}}) - \nu(g_{\text{ext}}) g_{\text{ext}}. \quad (2)$$

For $g_{\text{ext}} = 0$ this reduces to Eq. (1). *Using $\nu(y) = \sqrt{1 + a_0/y}$ in both Eqs. (1) and (2)—rather than borrowing the unrelated “simple” interpolation $\nu = \frac{1}{2} + \sqrt{\frac{1}{4} + a_0/y}$ for the EFE—is the single change that makes the theory use one law. It is required for internal consistency: a modified-gravity/inertia theory has one interpolation, fixed by its dynamics, not two chosen by convenience.*

The key limit. Deep inside a system ($g_N \ll g_{\text{ext}}$) in a weak external field, expand Eq. (2) to first order in g_N :

$$g_{\text{obs}} \simeq g_N \nu(g_{\text{ext}}) \left[1 + \frac{d \ln \nu}{d \ln g_{\text{ext}}} \right] \equiv g_N \frac{G_{\text{eff}}}{G}, \quad (3)$$

with, for the de Sitter–Unruh ν , the closed form $d \ln \nu / d \ln g_{\text{ext}} = -\frac{1}{2} [1 + g_{\text{ext}}/a_0]^{-1}$. The internal dynamics are Newtonian but with a constant G_{eff} set by the *external* field alone. As $g_{\text{ext}} \rightarrow 0$ the full boost returns; as $g_{\text{ext}} \gg a_0$, $G_{\text{eff}} \rightarrow G$ and gravity becomes exactly Newtonian. The external field *erases* the boost.

III. THE PREDICTION, IN PICTURES

Figure 1 plots the RAR from Eq. (2) on the framework’s de Sitter–Unruh ν for an isolated galaxy and for

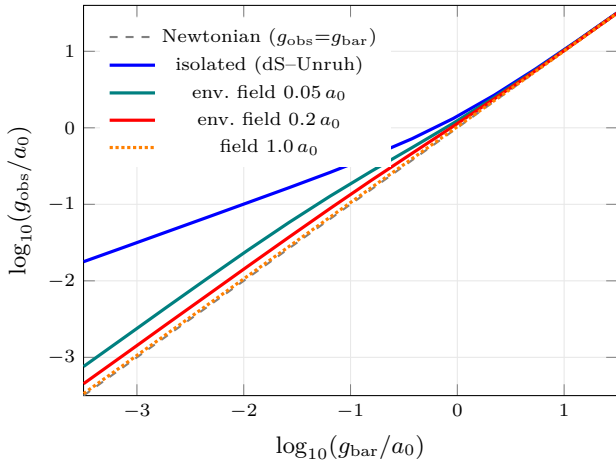


FIG. 1. The radial acceleration relation with the external field effect, on the framework’s de Sitter–Unruh interpolation used for *both* the isolated RAR and the EFE. An isolated galaxy (blue) lifts along the deep-MOND slope $\frac{1}{2}$; a galaxy in an environmental field bends to a plateau parallel to Newton, raised by $\nu(g_{\text{ext}})$ and lowered as the field strengthens. The dependence on the *environment* is forbidden for Newton+dark matter.

environmental fields of $0.05 a_0$, $0.2 a_0$, and $1.0 a_0$. The isolated curve (blue) follows the deep-MOND slope $\frac{1}{2}$; a galaxy in an external field bends to a *plateau* parallel to Newton once $g_N < g_{\text{ext}}$, raised by $\nu(g_{\text{ext}})$ and lowered as the field strengthens. This environment-dependent flattening is the EFE’s signature and is impossible for Newton+dark matter.

Figure 2 isolates the magnitude: the boost $g_{\text{obs}}/g_{\text{bar}}$ deep in the MOND regime ($g_{\text{bar}} = 0.01 a_0$) as a function of the external field. With no external field the boost is ~ 10 ; a Milky-Way-like field, which on the framework’s a_0 sits at $g_{\text{ext}} = 2.30 a_0$, cuts it to ≈ 1.0 along the field axis. The orbit-averaged boost for a wide binary in that field is $G_{\text{eff}}/G \simeq 1.14$ (+6.6% in velocity)—between zero (dark matter) and the isolated value, and *smaller* than the ~ 1.3 of the simple interpolation precisely because the lower a_0 deepens the suppression.

IV. WHERE THE CORRECTED LAW STANDS AGAINST DATA

The qualitative prediction is unchanged and remains a clean fork: galaxies in stronger external fields should show a more pronounced low-acceleration downturn in their RAR, which dark matter forbids. The *quantitative* numbers, computed honestly on the framework’s own law, are more modest than the simple-interpolation versions and must be reported as such:

(i) *SPARC environmental RAR.* An in-house test on SPARC with a real 2MRS external-field map, using the de Sitter–Unruh ν throughout, gives a kinematic–environmental correlation that leans the predicted (posi-

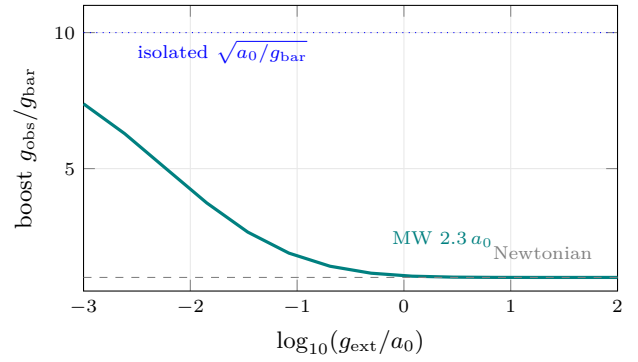


FIG. 2. The boost deep in the MOND regime ($g_{\text{bar}} = 0.01 a_0$) collapses from ~ 10 (isolated) to 1 (Newtonian) as the external field strengthens past a_0 , on the de Sitter–Unruh interpolation. The Milky Way’s field sits at $2.3 a_0$ on the framework’s a_0 .

tive) way in the constrained subsample but is not significant ($r \simeq +0.21$, $p \simeq 0.2$); the binding limit is the small dynamic range of the cosmic-web field across the SPARC sample, not the interpolation. This is neither a confirmation nor a refutation. The external-field detections reported in the literature [3] are at ~ 4 – 5σ but remain contested.

(ii) *Wide binaries.* Stars at separations $\gtrsim 5000$ AU have internal accelerations below a_0 while sitting in the Galaxy’s $2.30 a_0$ field. On the de Sitter–Unruh law the predicted internal boost is $G_{\text{eff}}/G = 1.20$ (transverse) and 1.02 (along the field), orbit-averaging to 1.14—a velocity excess of +6.6% over Newton, *below* the $\sim +15\%$ a simple interpolation would give. This is a clean “is-gravity-modified-at-all” discriminator against Newton (the excess is well above zero) but is degenerate with standard MOND in the value of a_0 , and the present measurements straddle: it sits below the boosted central values reported by some analyses and consistent with the Newtonian-baseline reanalyses. Gaia DR4 will sharpen, though not by itself resolve, the value of a_0 [4].

We state plainly: the corrected, internally-consistent law predicts a *smaller* environmental signal than the mixed-interpolation version, and the correct response to the data today is “consistent, not confirmed.”

V. WHERE a_0 COMES FROM—AND WHAT IS NOT DERIVED

The scale a_0 is inherited from cosmology. In a universe with a cosmological constant $\Lambda > 0$ the de Sitter horizon has curvature $\sqrt{\Lambda}$, and the emergent acceleration it sets is

$$a_0 = c^2 \sqrt{\Lambda/32\pi} = cH_\Lambda/Z, \quad Z = 2\sqrt{8\pi/3}, \quad (4)$$

the coincidence $a_0 \approx cH_0$ made precise: Eq. (4) gives $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$ for the measured Λ . In this

reading the EFE is the *non-locality of the horizon’s coupling*: the modified inertia descends from the global de Sitter geometry, so the field an object sits in—set partly by its neighbours—enters its internal physics. Dark energy (Λ) and the “dark matter” boost are two faces of one geometry. We flag honestly that the coefficient $Z = 2\sqrt{8\pi/3} = \sqrt{32\pi/3} = 5.79$ is an *unforced posit*: it is not derived here, and a_0 remains hostage to H_0 at the $\mathcal{O}(1/6)$ level. The distinctive, falsifiable content of the framework is the predicted decline $a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}$, in which Z cancels.

VI. CONCLUSION

The external field effect is the cleanest fork in the dark-sector road: Newton+dark matter cannot make a system’s internal dynamics depend on a uniform external field; a non-linear modified-inertia law must. We

have written that law with *one* interpolation—the framework’s own de Sitter–Unruh $\nu(y) = \sqrt{1 + a_0/y}$ for both the isolated RAR and the EFE—removing the internal inconsistency of mixing it with the simple MOND interpolation. The corrected predictions are sharper and, honestly, smaller: a wide-binary boost of +6.6%, not +15%, because the framework’s lower a_0 deepens the external-field suppression. The qualitative discriminator stands undefeated so far—a *galaxy’s gravity knows about its neighbours*—while the quantitative value of a_0 , and the test of its predicted redshift decline, await Gaia DR4 and high- z kinematics.

Note. This paper is the de Sitter–Unruh-consistent companion to an earlier EFE treatment by the same author that used the simple MOND interpolation for the external-field law; the earlier paper is preserved as-is. The figures and boost numbers here supersede it wherever the two differ.

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